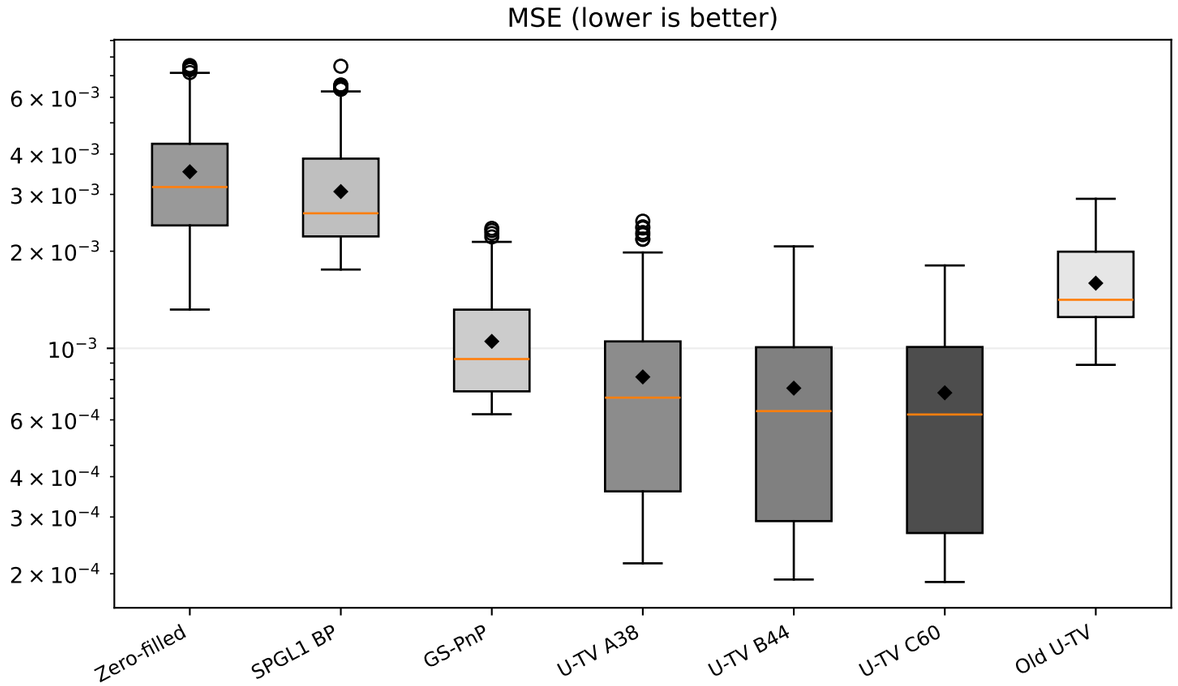
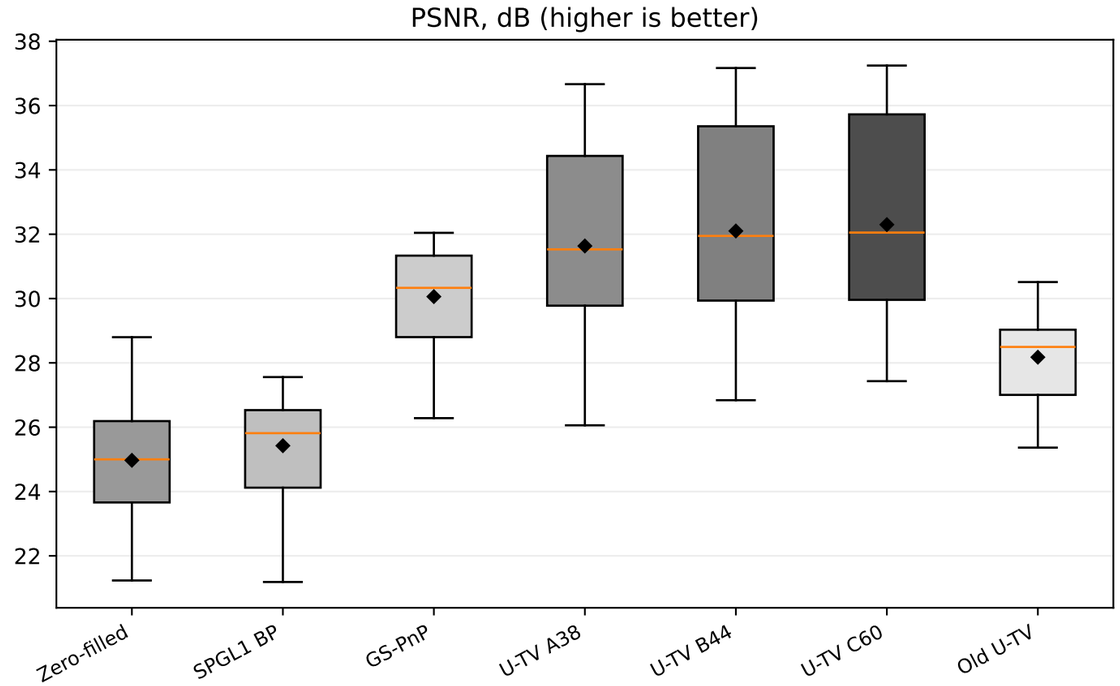
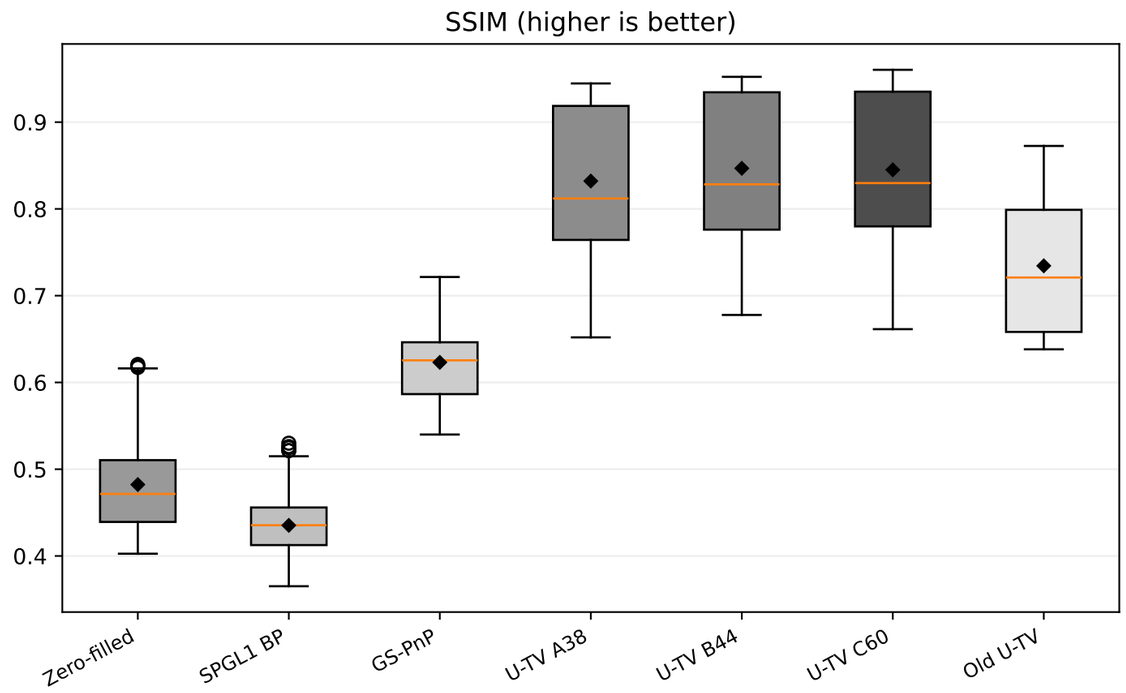
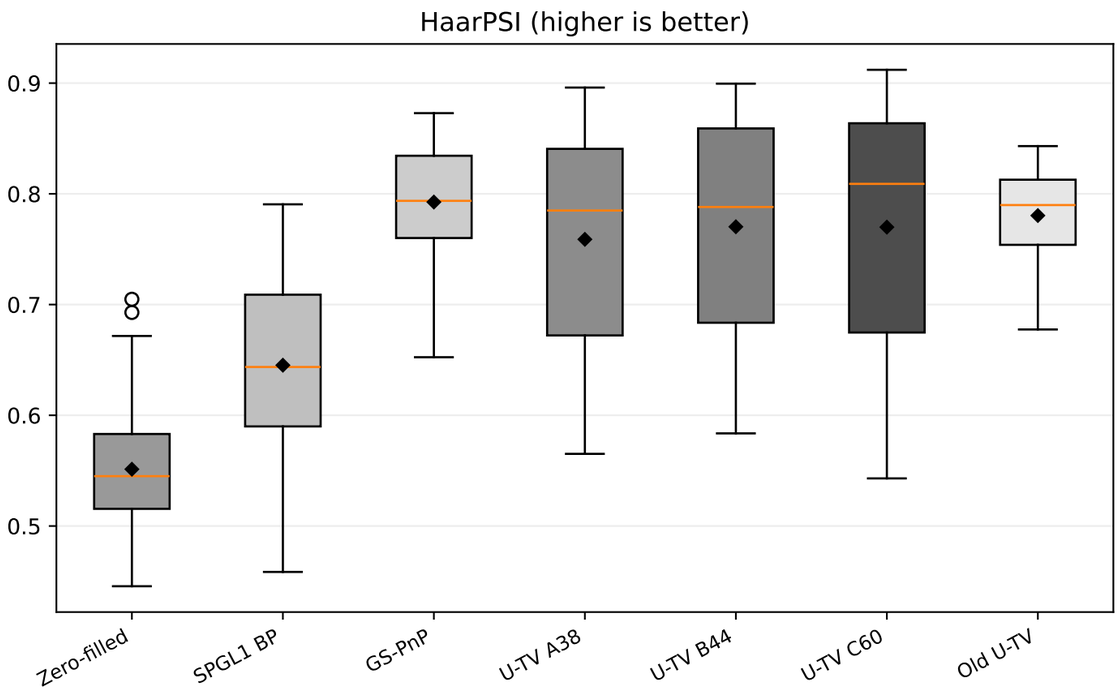
# All 450 test images | fixed 20% | NPL default noise



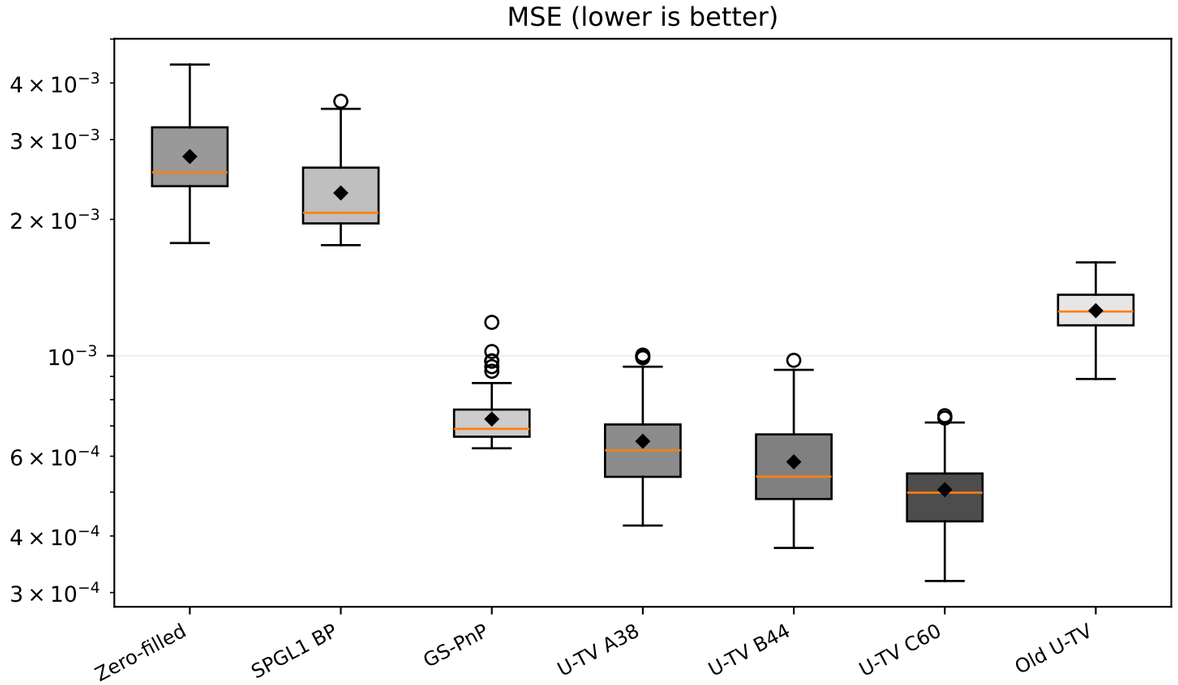



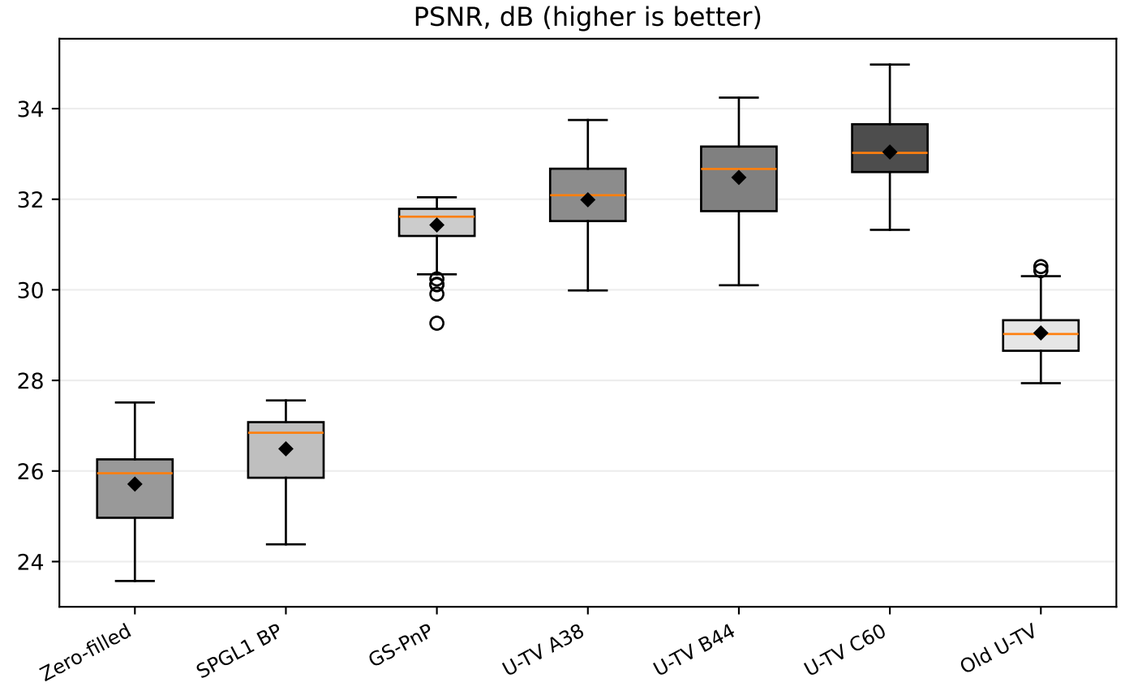



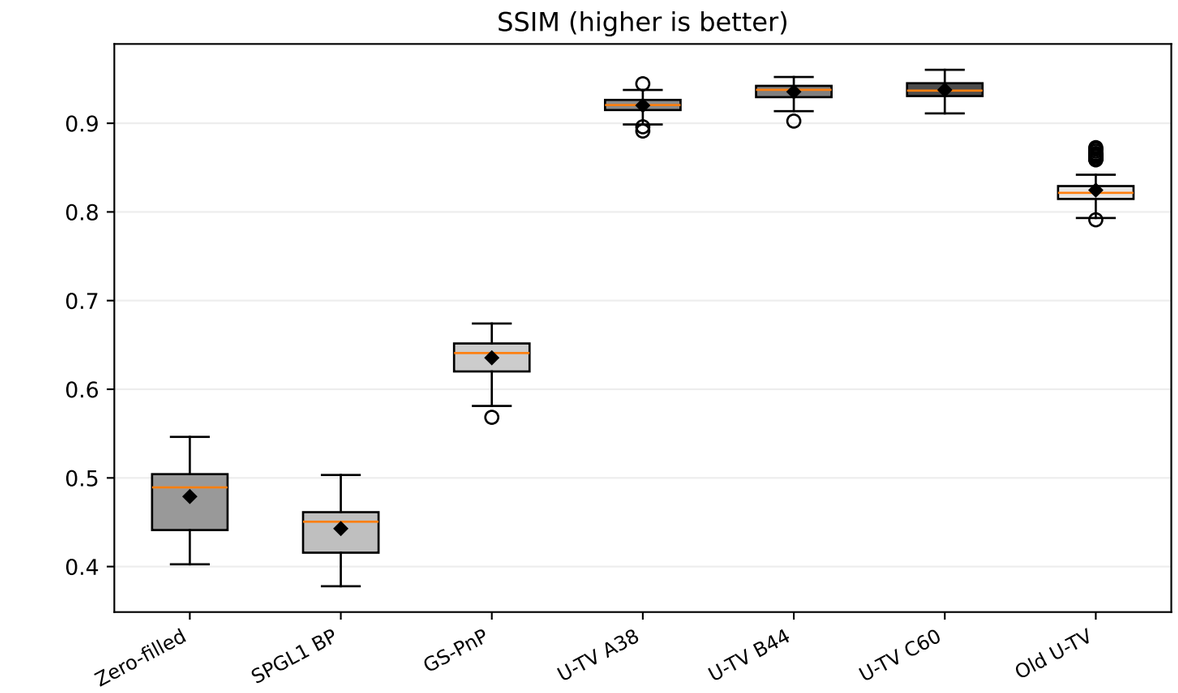



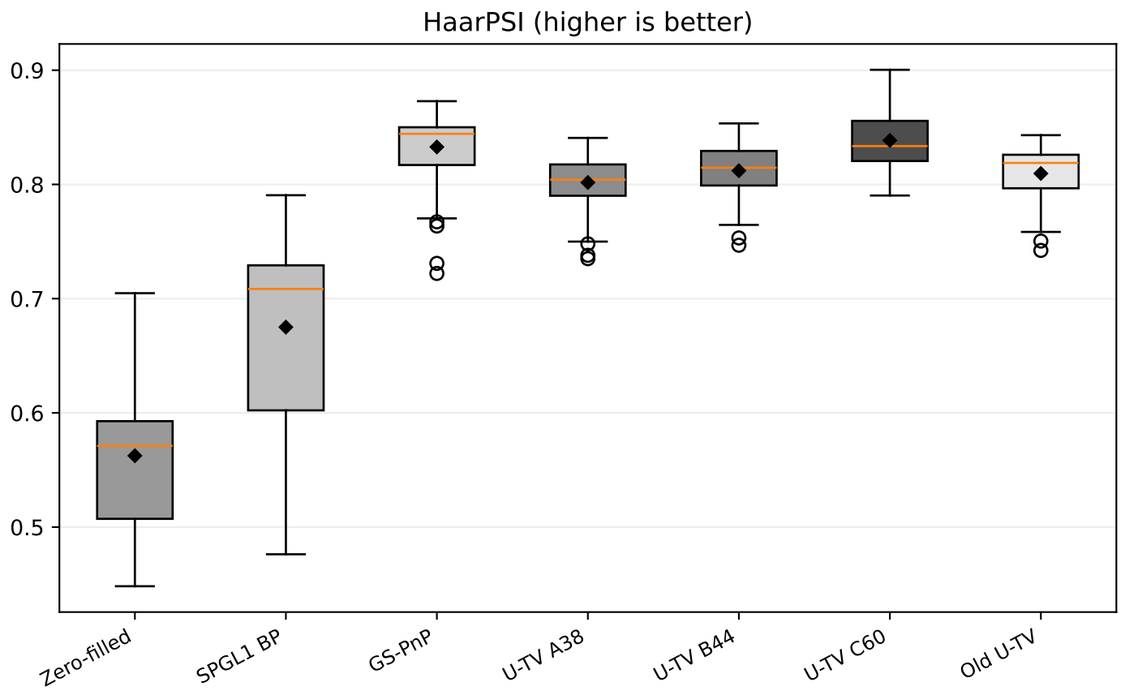

Each box has 450 paired test images; median, IQR, 1.5 x IQR whiskers and outliers. Black diamond: mean. MSE uses log scale.

# CIGS: 90 held-out test images | fixed 20% | NPL default noise



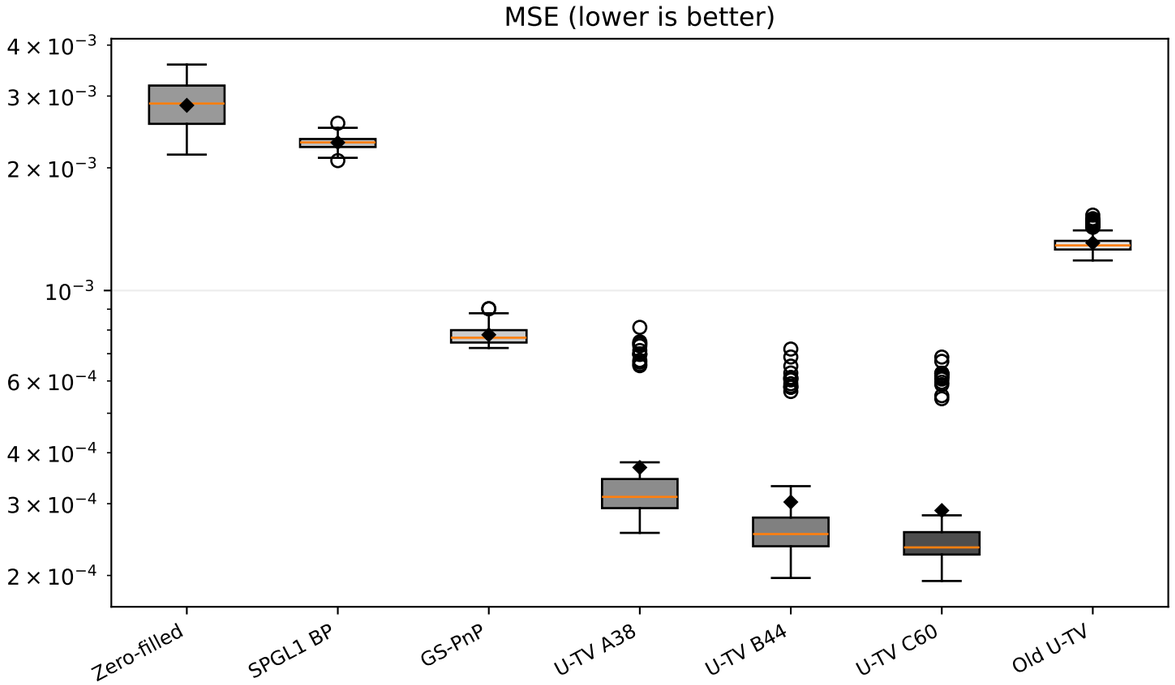



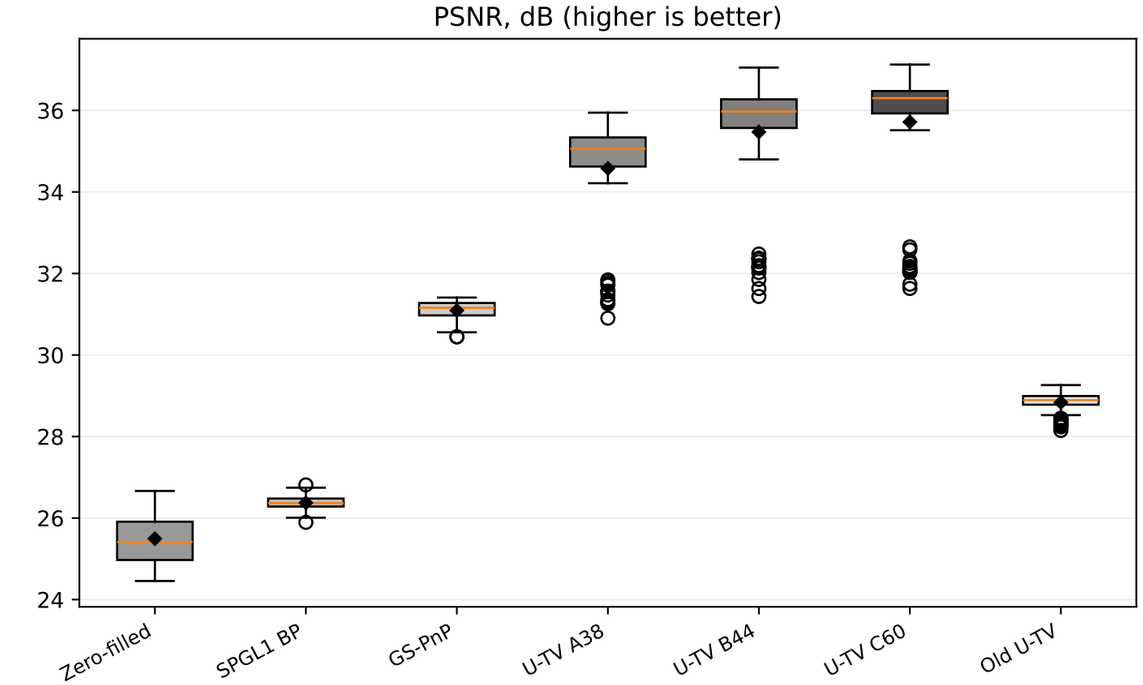



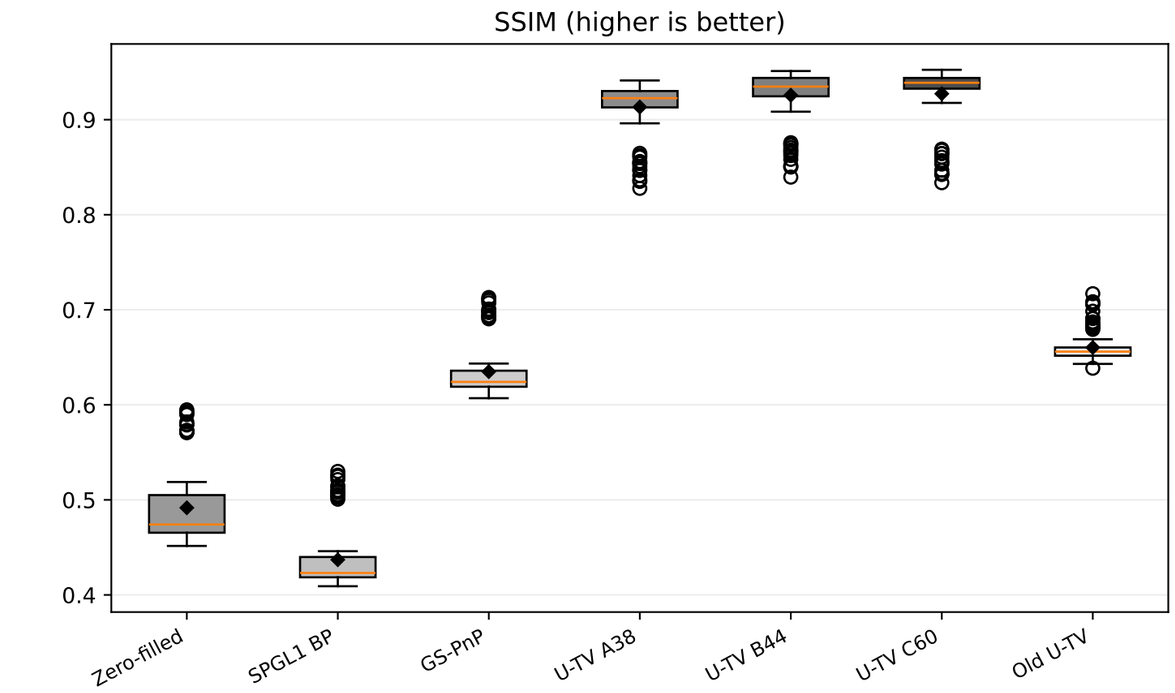



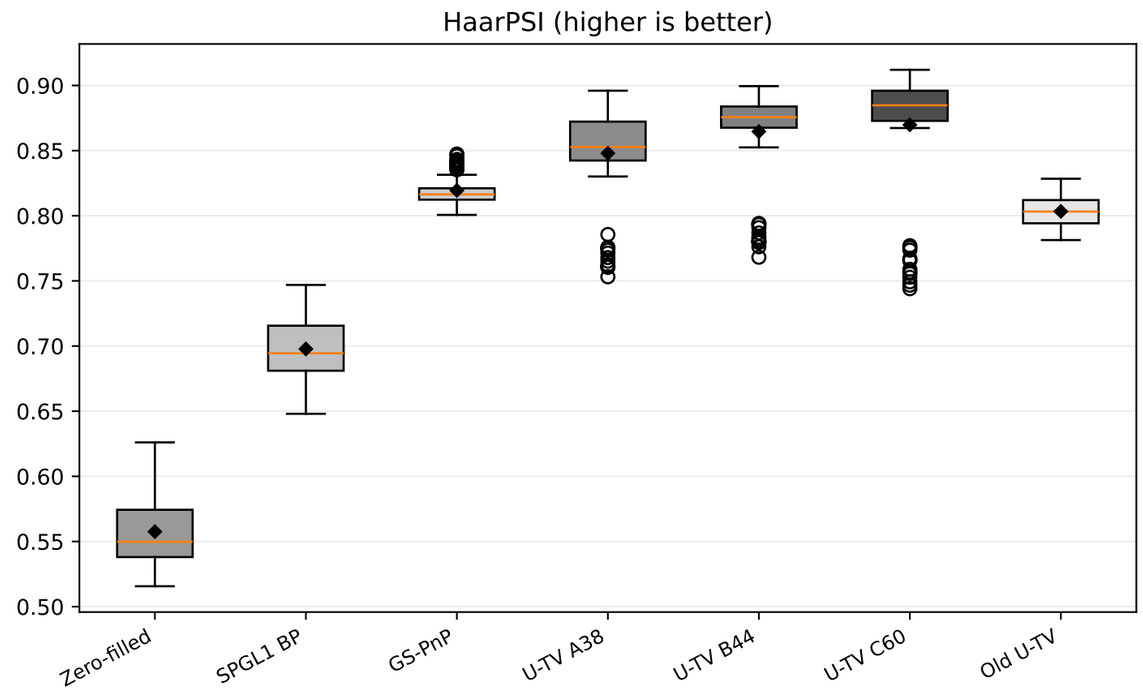

Each box has 90 paired test images; median, IQR, 1.5 x IQR whiskers and outliers. Black diamond: mean. MSE uses log scale.

# Organic: 90 held-out test images | fixed 20% | NPL default noise



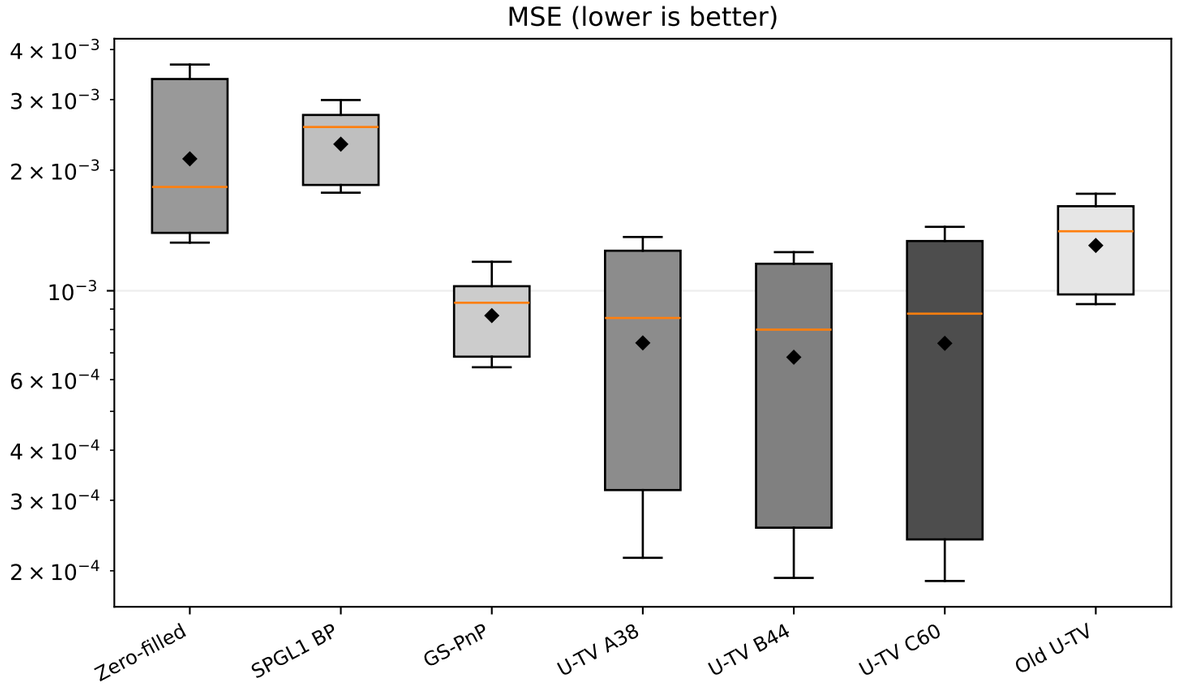



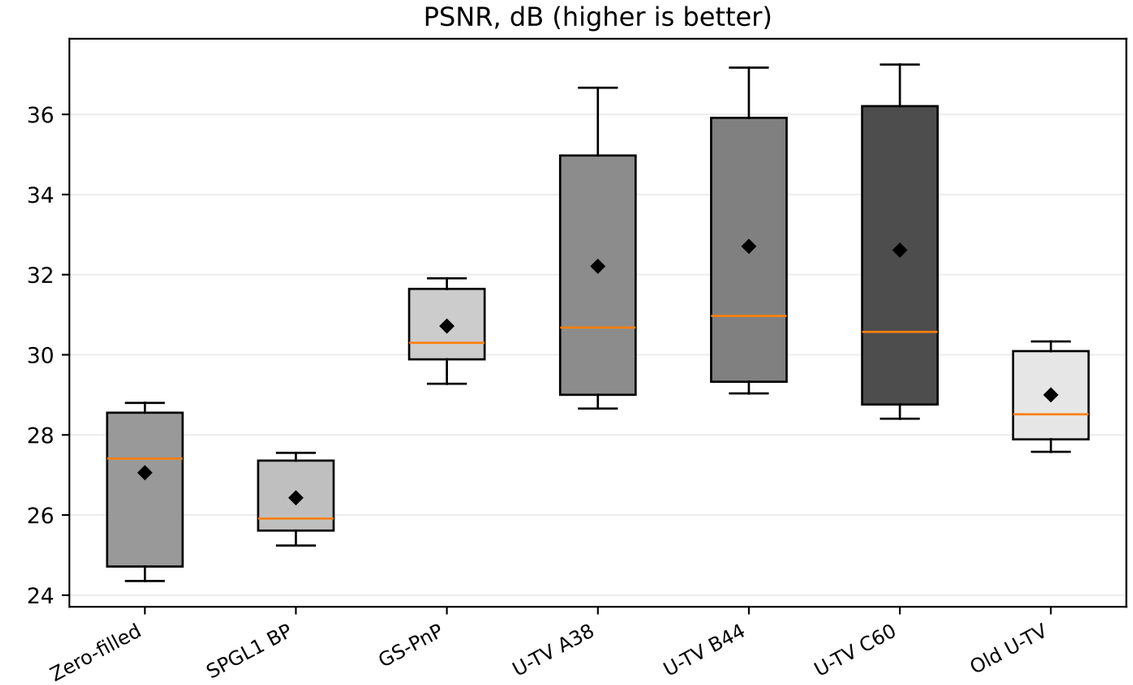



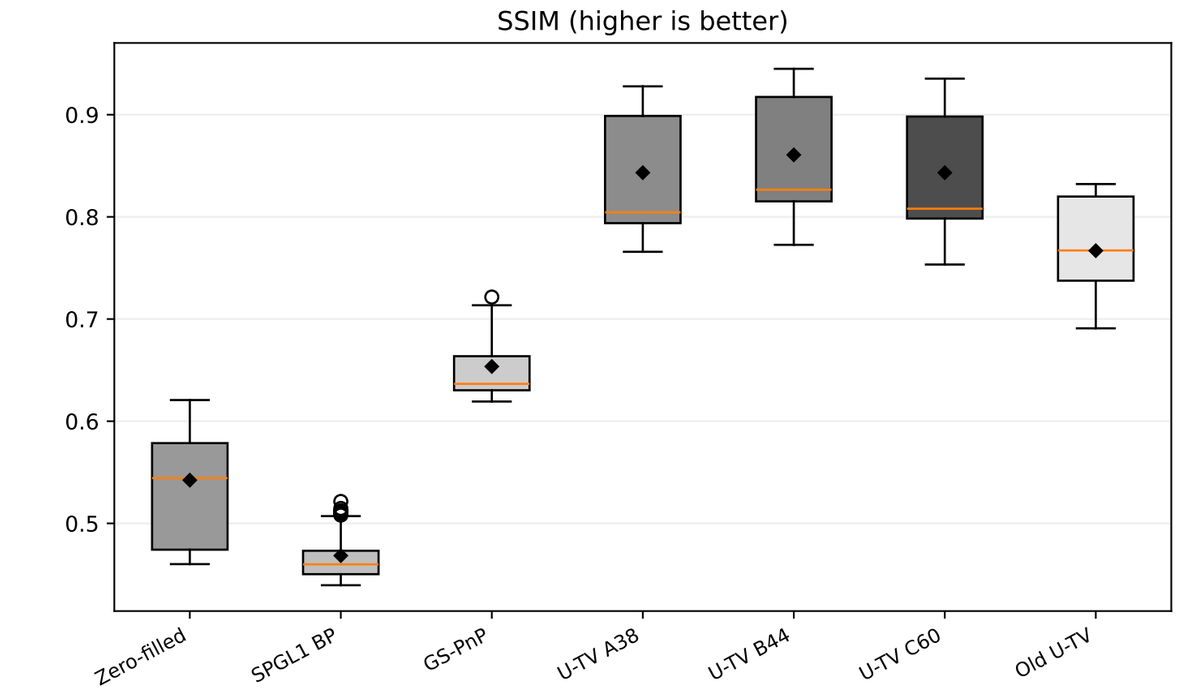



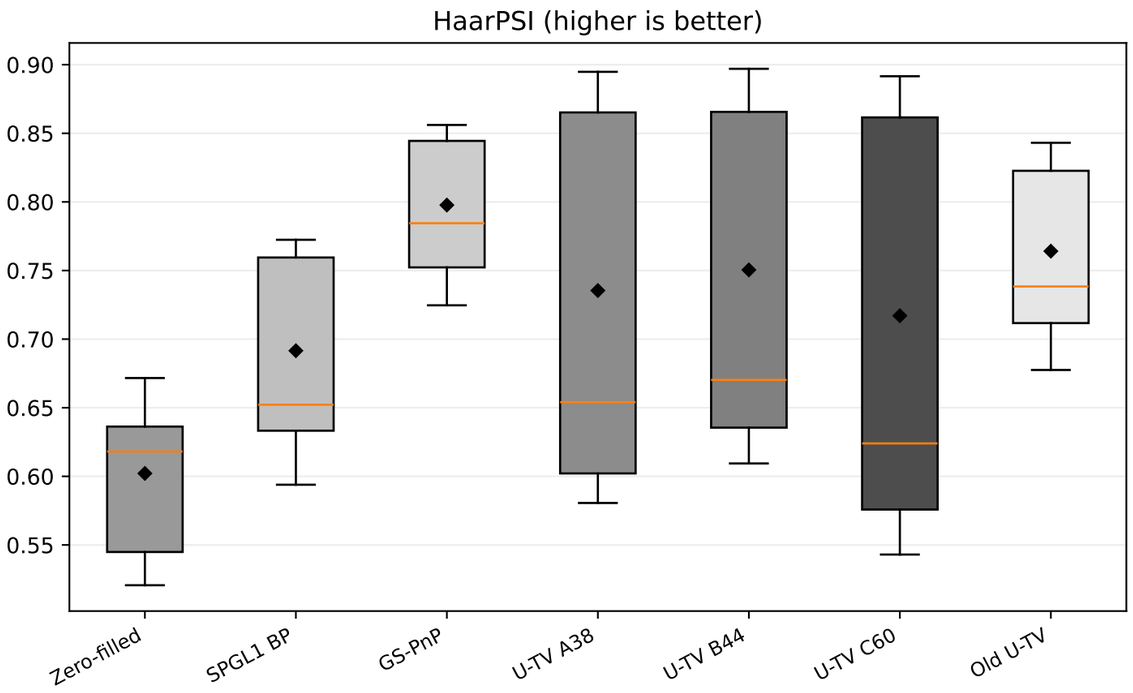

Each box has 90 paired test images; median, IQR, 1.5 x IQR whiskers and outliers. Black diamond: mean. MSE uses log scale.

# Perovskite: 90 held-out test images | fixed 20% | NPL default noise



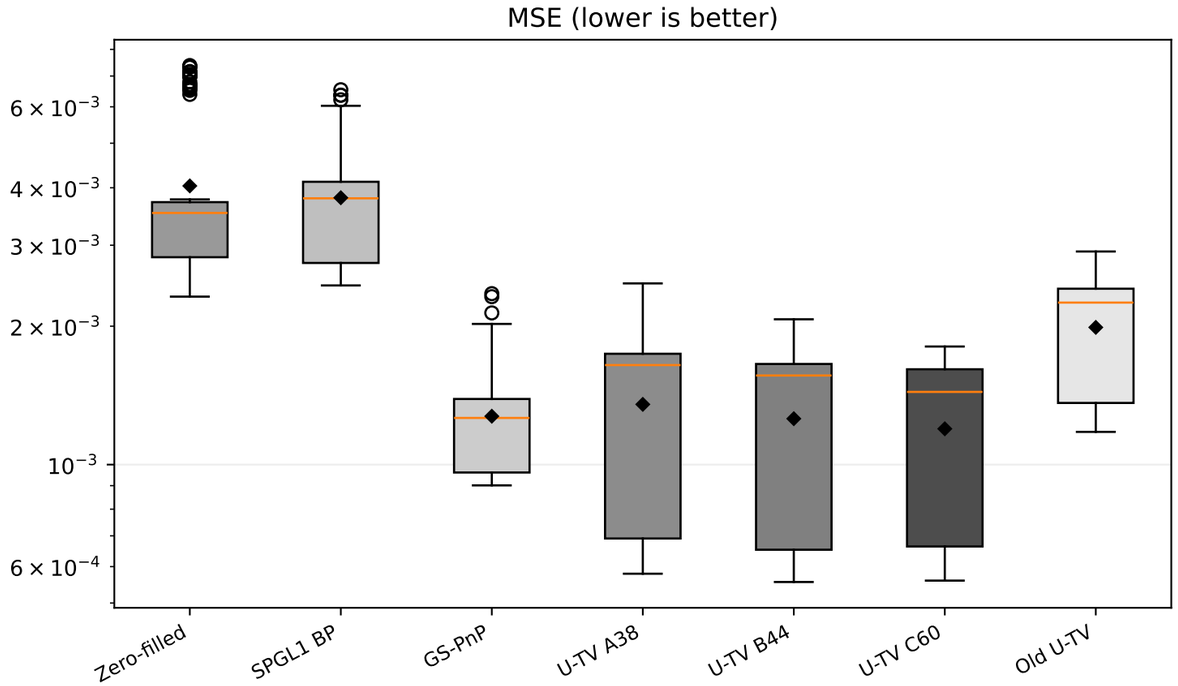



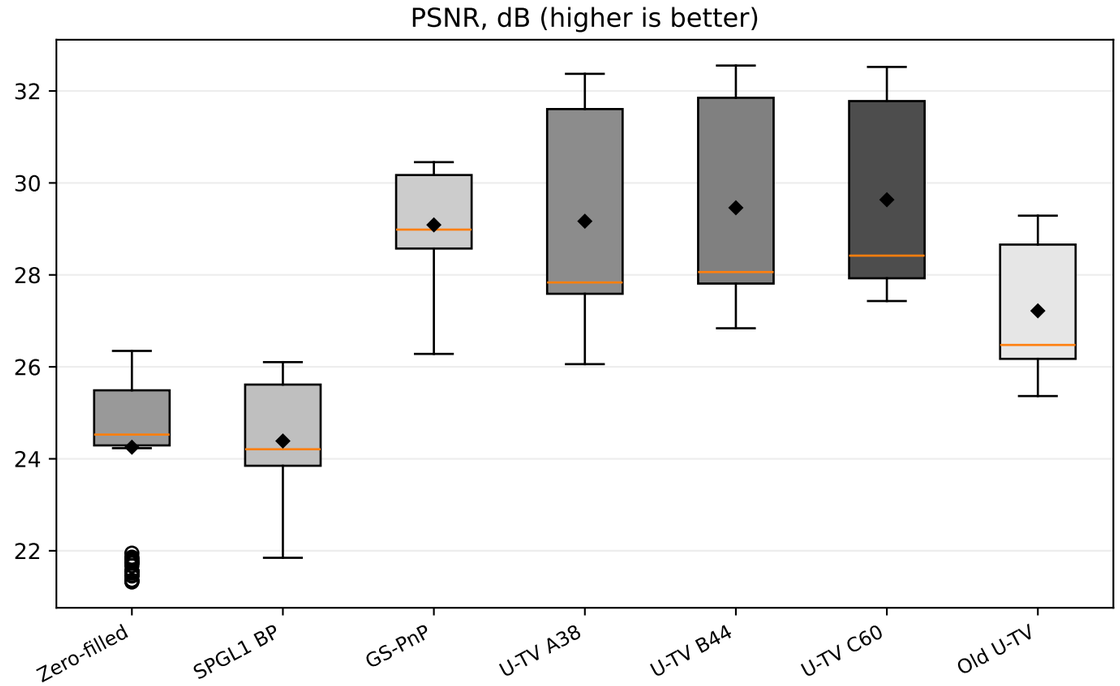



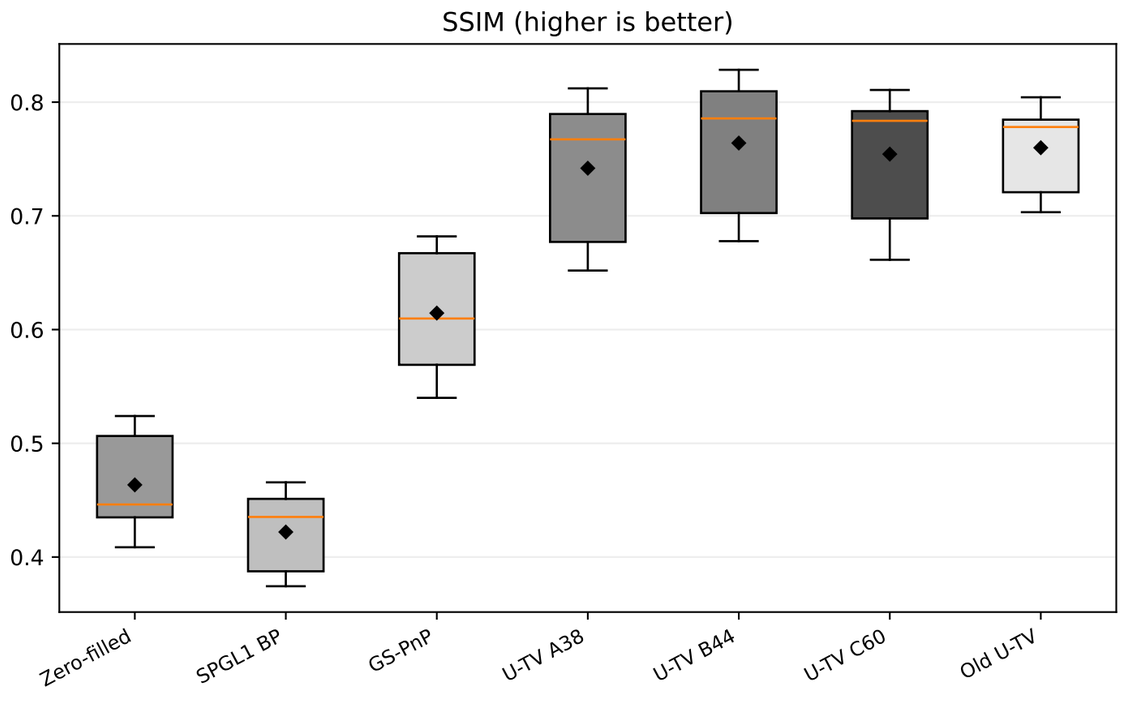



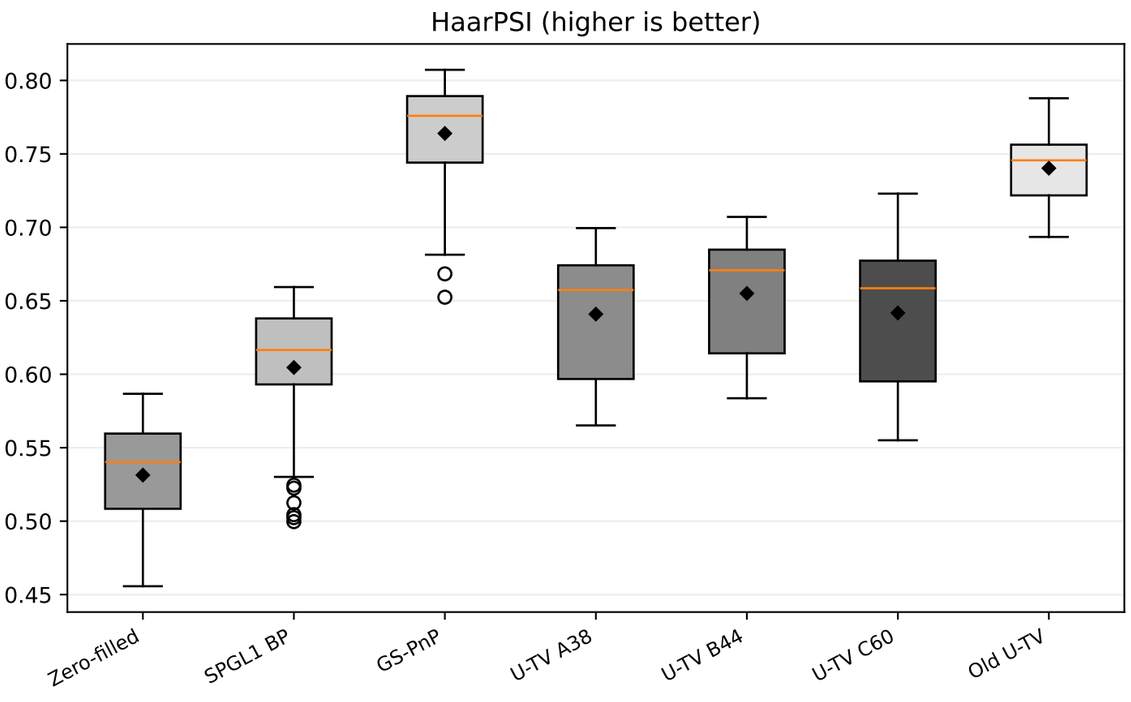

Each box has 90 paired test images; median, IQR, 1.5 x IQR whiskers and outliers. Black diamond: mean. MSE uses log scale.

# Reference: 90 held-out test images | fixed 20% | NPL default noise



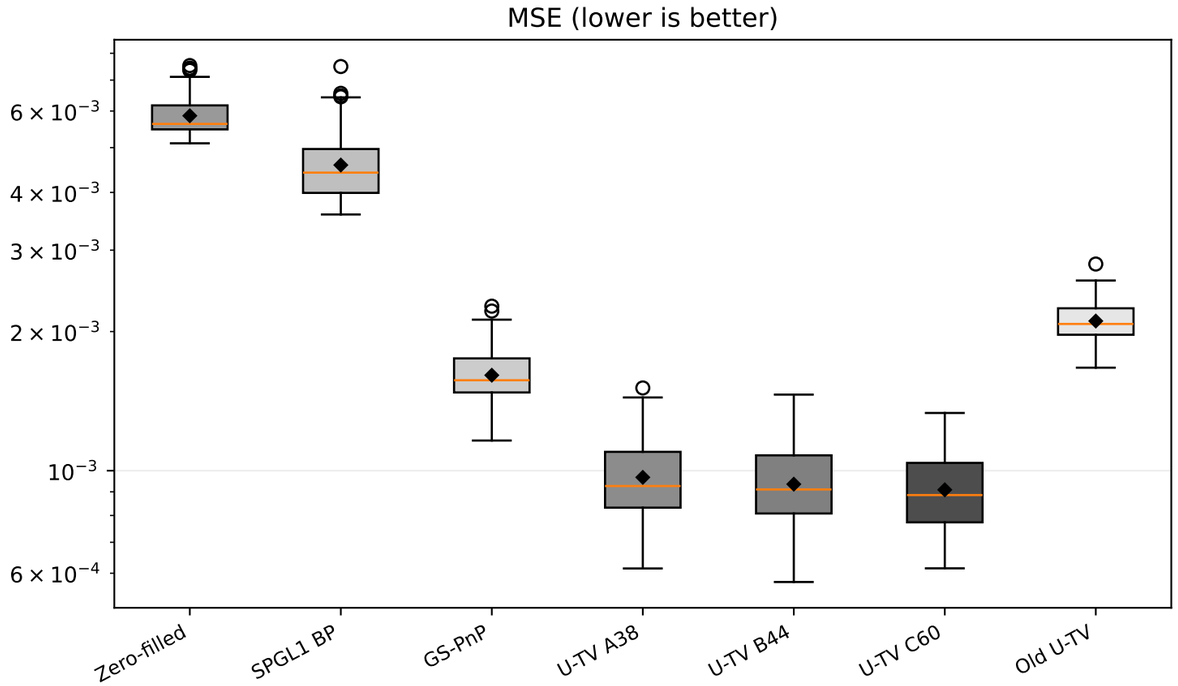



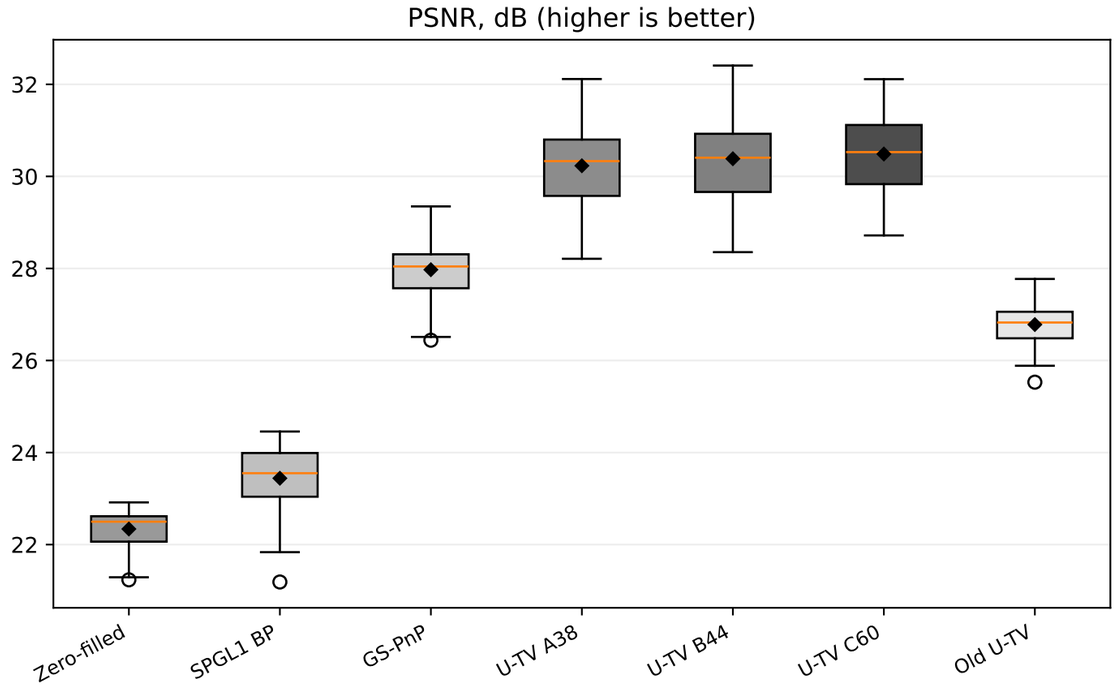



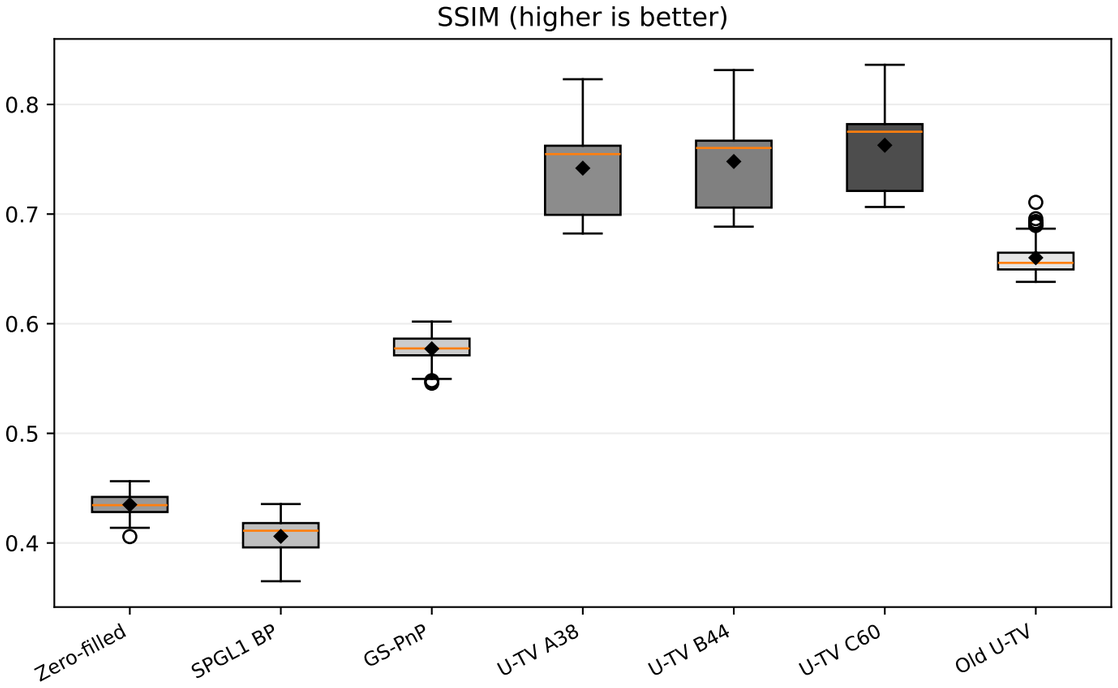



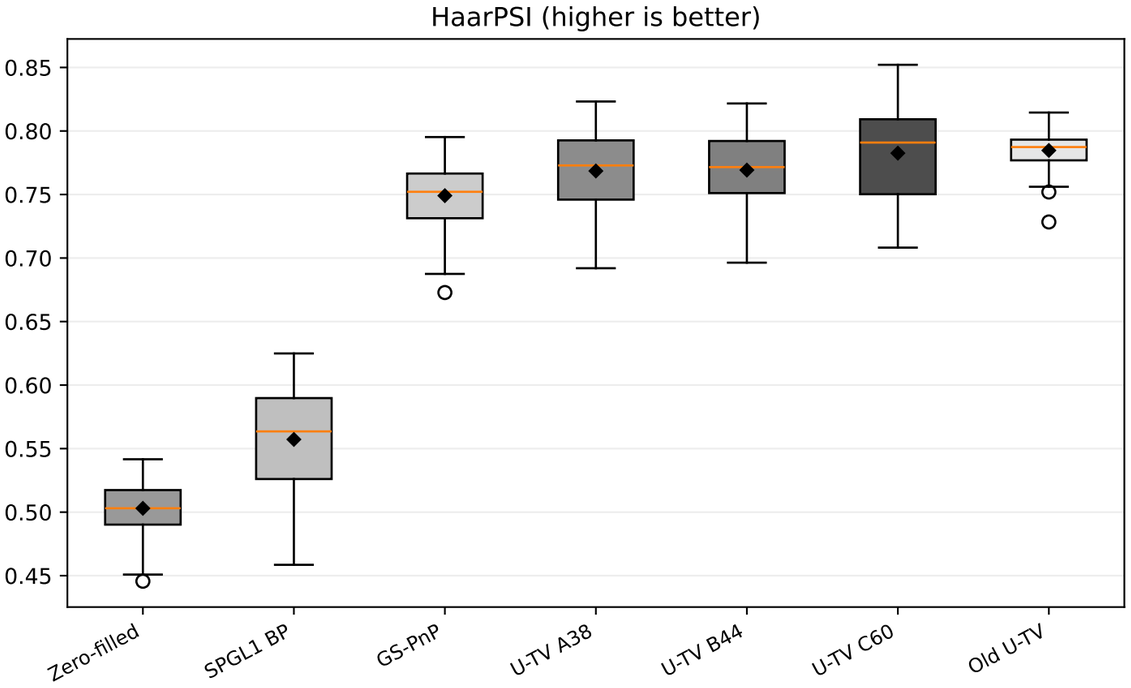

Each box has 90 paired test images; median, IQR, 1.5 x IQR whiskers and outliers. Black diamond: mean. MSE uses log scale.

# Silicone: 90 held-out test images | fixed 20% | NPL default noise




Each box has 90 paired test images; median, IQR, 1.5 x IQR whiskers and outliers. Black diamond: mean. MSE uses log scale.

# Boxplot protocol and limitations

01. Five categories, each with 90 held-out synthetic PCM images.

02. Sampling ratio = 0.20 for every image, multilevel Hadamard, J=8, coarse J=6.

03. Every acquisition uses the same NPL-default noise parameters; no sigma sampling.

04. Independent noise traces and masks are deterministically seeded per image.

05. All seven methods see exactly the same sampled operator and noisy measurement.

06. Original SPGL1 is basis pursuit (db4, max 100 iterations), then 10 debias steps.

07. GS-PnP is the pretrained GS-DRUNet preset with lambda=0.1, sigma=0.05, max 50 iterations.

08. U-TV A epoch 38, B epoch 44, C epoch 60, old clean U-TV epoch 99.

09. The old and new U-TV models also differ in architecture and unrolling depth.

10. The full test contains one noise realisation per image; this cannot measure repeated-noise variability.

11. MSE uses a log axis so severe outliers remain visible; the box diamonds show means.

12. Synthetic images with simulated measurements are not physical measured PCM acquisitions.